

# SASHIDHAR MOTTE

Buffalo, New York

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## SUMMARY

Full-stack engineer who builds and ships internal tools and data-heavy dashboards used daily. Built and operated an internal CX platform handling 10–15K queries/day (React frontend through APIs, auth, and payments). Experienced building agentic LLM systems, RAG pipelines, model evaluation, and end-to-end ML features; comfortable driving projects with ambiguous requirements and partnering with teams to measure impact.

## EXPERIENCE

### University at Buffalo

October 2025 – Present

*Research Assistant - Software/AI Engineer*

*Buffalo, NY*

- Cross-Platform Desktop App: Built a full-stack Tauri photo application (Apple Photos-style) with a React/TypeScript frontend and native backend, designing a fast, responsive gallery and natural-language search over thousands of photos.
- Custom Text Encoder Training: Trained a lightweight text encoder on synthetically generated data from photo metadata, improving search recall from 62% to 92% on a personal photo dataset.
- On-Device ML Pipeline: Ran the full search pipeline on device under 6MB via quantization and a hybrid cloud-device sync strategy, keeping the interface fast under real constraints.
- Knowledge Graph for Memory Surfacing: Explored knowledge graphs to automatically surface interesting photo memories, connecting people, places, and activities across a gallery.

### Aspora (YC W22)

November 2023 – November 2024

*Full Stack Engineer*

*Bangalore*

- Data-Rich Dashboard (Alphadesk): Built and owned an internal dashboard end to end — React/TypeScript interface plus backend APIs — used by the CX team daily to handle customer issues, payments, and KYC/authentication, serving 10–15K queries/day.
- Observability & Data Visualization: Built SQL and Mixpanel dashboards that made large-scale product data intuitive, surfacing KPIs, funnel conversion, and latency so teams could act on them.

## PROJECTS

### Agentic LLM: Tool Use, RAG & Evaluation

- Built an agentic LLM system with tool use and a hybrid RAG pipeline (e5-base-v2) that decides when to fetch evidence before acting, grounding outputs and cutting hallucination.
- Improved accuracy 3.3x (12% to 40%+) via GRPO fine-tuning implemented from scratch (no critic network).
- Designed the evaluation framework to measure output quality and catch regressions.

### DarkWatch: Data-Heavy Observability Dashboard

- Designed and built a React dashboard with complex data visualization, including an interactive 3D before/after globe that makes large-scale change data intuitive at a glance.
- Owned the full stack behind it: the FastAPI API, the data-processing pipeline, and deployment, taking the product from raw data to production.
- Trained a Siamese ResNet + UNet model (F1 0.59 to 0.80, +35%) and evaluated it on 7,120 labeled samples to validate the results the dashboard surfaces.

## TECHNICAL SKILLS

**Frontend:** React, Next.js, TypeScript, Redux, TailwindCSS, Vite, Rollup, Tauri, Complex UI, Data Visualization, Responsive/Accessible UI

**Full-Stack & Data:** Node.js, Express, FastAPI, Java/Spring Boot, REST API Design, **PostgreSQL**, MongoDB, Redis, Authentication, Data Modeling

**Observability:** Dashboards, Analytics, Mixpanel, SQL/Metabase, Monitoring, Performance

**AI & Agents:** LLMs, Agentic Workflows, RAG, Tool Use, Evaluation, AI Coding Tools, PyTorch, HuggingFace

**Infra:** AWS, Docker, CI/CD, Git, Linux

## EDUCATION

### State University of New York, Buffalo

2025 – 2026

*MS CS(AI/ML) - CGPA - 3.8/4*

*Buffalo, New York*

### Indian Institute of Information Technology, Sri City

2019 – 2023

*B. Tech. (CSE) - CGPA - 8/10*

*Sricity, India*